INCH-POUND

MIL-DTL-32554B
6 August 2020
SUPERSEDING
MIL-DTL-32554A
28 June 2018

DETAIL SPECIFICATION

STRAPS, TIE-DOWN, ADJUSTABLE, NON-METALLIC, CABLE BUNDLING, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design and test requirements for adjustable, non-metallic, cable bundling tie-down straps primarily intended for Electrical Wiring Interconnect System (EWIS) secondary support including fiber optic and co-axial cable within aircraft applications.

1.2 Part or Identifying Number (PIN). The PINs to be used for the tie-down straps acquired in accordance with this specification are created as follows:

M32554 – XX –X –XXX

- XXX: Color (see Table I)
- X: Lacing tape material designator (see Table I)
- XX: Strap length designator (see Table II)
- 32554: Detail specification number
- M: M prefix

Comments, suggestions, or questions on this document should be addressed to Commanding Officer, Naval Air Warfare Center Aircraft Division Lakehurst, Systems Standardization and PHS&T Branch, Code BL32600, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100 or emailed to frank.magnifico@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at https://assist.dla.mil.

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1.3 Temperature rating. -65 °C to +260 °C (-85 °F to +500 °F).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This document does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATION

TT-I-735	-	Isopropyl Alcohol

COMMERCIAL ITEM DESCRIPTION

A-A-52084	-	Tape, Lacing and Tying, Aramid

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5606	-	Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance
MIL-PRF-7808	-	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-DTL-32555	-	Tool, Installation, for Adjustable Tie-Down Straps, General Specification for
MIL-DTL-83133	-	Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO F-34), NATO F-35, and JP-8+100 (NATO F-37)
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric, NATO Code Number H-537
MIL-PRF-87257	-	Hydraulic Fluid, Fire Resistant; Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-202	-	Electronic and Electrical Component Parts
MIL-STD-202-107	-	Method 107, Thermal Shock
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM D1655	-	Standard Specification for Aviation Fuels
ASTM D5025	-	Standard Specification for Laboratory Burner Used for Small-Scale Burning Tests on Plastic Materials

(Copies of these documents are available from http://www.astm.org.)

NATIONAL CONFERENCE OF STANDARDS LABORATORIES (NCSL) INTERNATIONAL

NCSL Z540.3	-	Requirements for the Calibration of Measuring and Test Equipment

(Copies of this document are available from http://www.ncsli.org.)

SAE INTERNATIONAL

SAE AIR1351	-	Manufacturers' Identification of Aerospace Electrical and Electronic Wiring Devices and Accessories
SAE AS21919	-	Clamp, Loop Type, Cushioned Support
SAE AS22759/10	-	Wire, Electrical, Fluoropolymer-Insulated, Extruded TFE, Nickel-Coated Copper Conductor, 1,000 Volt, ROHS
SAE AS22759/16	-	Wire, Electric, Fluoropolymer-Insulated, Extruded ETFE, Medium Weight, Tin-Coated Copper Conductor, 600-Volt, 150 °C, ROHS

(Copies of these documents are available online at http://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The tie-down straps furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified product list before contract award (see 4.3 and 6.3).

3.2 Tie-down straps. The tie-down straps shall consist of a length of cable lacing tape, permanently and mechanically attached to a fastener.

3.3 Materials. The materials used in the manufacture of the tie-down straps shall be such

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that the tie-down straps are capable of meeting all of the performance and test requirements of this specification. The lacing tape used in the tie-down strap assembly shall be as specified in Table I. No metal may be used in the construction of the tie-down strap assembly.

3.4 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.5 Design and construction. The design and construction of the tie-down straps shall be as specified herein. Design shall conform to all performance requirements in 3.7. The design allows for installation of tie-down straps using single, double or triple wrapping of lacing tape around cable bundle before securing as shown on Figure 1. The tie-down strap assembly configuration is optional within the maximum dimensions (see Figure 2 and Table II).

3.5.1 Fastener design. Fastener shall be designed to accommodate wrapping of tie-down straps around cable bundle, threading of lacing tape through fastener lacing tape path, tensioning of tie-down straps around cable bundle, and securing tie-down straps to cable bundle. The tie-down straps shall be secured on cable bundle with automatic activation when the installer applies 17 to 22 pounds-force (lbf) tensile force to the free end of the lacing tape (see Figure 2).

3.6 Dimensions and tolerances. Dimensions for the various sizes of tie-down straps are as shown in Table II.

3.7 Performance. The tie-down strap assembly shall conform to the requirements of this specification, when subjected to the applicable tests specified in Table III.

3.7.1 Tensile strength (see 4.6.1). The tie-down strap assembly shall meet the minimum tensile strength requirements of Table IV without breaking or releasing the tie-down strap from the tensile test fixture after being subjected to the tensile strength test of 4.6.1.

3.7.2 Thermal shock. The tie-down strap assembly shall not deform, crack, break or release from the cable bundle, when subjected to the thermal shock test of 4.6.2. Following the thermal shock test, the tensile strength after temperature exposure test (see 4.6.4) shall be performed. The tie-down straps shall meet the minimum tensile strength requirements of Table IV after the thermal shock test.

3.7.3 Temperature life cycle. The tie-down strap assembly shall not deform, break or release from the cable bundle, when subjected to the temperature life cycle test of 4.6.3. The tie-down straps shall meet the minimum tensile strength requirements of Table IV after the temperature life cycle test.

3.7.4 Vibration. The tie-down strap assembly shall be exposed to the vibration test of 4.6.5 without cracking, breaking, or releasing from the wire bundle.

3.7.5 Fluid immersion (see 4.6.6). The tie-down strap assembly shall be immersed in the designated fluids of Table V without cracking, distorting, breaking, separation of fastener

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head from lacing tape, or releasing the locking device. The tie-down straps shall meet the tensile strength requirements of Table VI and identification of product and legibility requirements of 3.7.9 upon completion of the fluid immersion test.

WARNING

The fluids listed in Table V are highly toxic to skin, eyes and respiratory tract. They are also flammable. Eye and skin protection is required. Good general ventilation is normally adequate.

3.7.6 Fuel immersion life. The tie-down strap assembly shall be immersed in the designated fluids specified in 4.6.7 without cracking, distorting, breaking, separation of fastener head from lacing tape, or releasing the locking device. The tie-down strap assembly shall meet the tensile strength requirements of Table VI upon completion of the fuel immersion test.

3.7.7 Slip resistance. When subjected to the slip resistance test of 4.6.8, the tie-down strap assembly shall meet the minimum slip resistance load requirements of Table VII without breaking or releasing the tie-down strap from the mandrel.

3.7.8 Flammability. The tie-down strap assembly shall be tested in accordance with the flammability test of 4.6.9. The flame shall be self-extinguishing within a maximum of 3 seconds after removal of the external flame source from the activated tie-down strap.

3.7.9 Identification of product (see 4.6.10). The manufacturer name or trademark (see SAE AIR1351) shall be marked on the tie-down strap fastener (see figure 2). The location on the fastener is optional. The PIN, manufacturing company name, and Commercial and Government Entity (CAGE) Code shall be marked on the tie-down strap assembly packaging. Identification shall be examined in accordance with 4.6.10.

3.7.10 Color. The fastener shall be black in color and the strap shall be as defined in Table I.

3.7.11 Workmanship. Workmanship shall ensure that the fastener and straps are of uniform quality, free from defects that would affect their service, and shall be uniform in appearance without any mold flash.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a Qualification inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified herein, all inspections shall be made

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at ambient temperature and humidity as specified in MIL-STD-202 test conditions.

4.2.1 Test equipment and inspection facilities. Test and measuring equipment and inspection facilities of sufficient accuracy and quantity to permit performance of the required inspection shall be established and maintained by the contractor.

4.2.2 Tie-down strap assembly. The tie-down strap shall be installed on the cables or mandrels by the test activity using the M32555/01-01 tie-down strap tool.

4.3 Qualification inspection. Qualification inspection shall consist of examinations and tests specified in Table III. The qualifying activity shall perform or witness the particular tests as specified in Table III.

4.3.1 Qualification inspection process. For guidance on the qualification inspection process, see 6.3 and 6.3.1.

4.3.2 Qualification inspection samples. The sample size needed for each part number to be qualified shall be as shown in Table III. Different part number tie-down strap lengths may be done by similarity (see 6.3.3).

a. For the qualifying activity tests, 92 tie-down straps (82 for testing and 10 spares) for each part number shall be separately packaged and forwarded to the address designated in the letter of authorization. Sufficient wire conforming to SAE AS22759/10 and /16 (see 4.6.2, 4.6.3, and 4.6.5) shall be provided to perform the designated tests. Each sample shall be identified by marking each package with the following information:

Sample for qualification:
Specification MIL-DTL-32554 Straps, Tie-Down, Adjustable, Non-Metallic, Cable Bundling
Part number: M32554-XX-X-XXX
Name of supplier
Supplier's part number and Commercial and Government Entity (CAGE) Code
Submitted (date) under authorization (reference letter authorizing the inspection)

4.3.3 Retention of qualification. Retention of qualification inspections shall be performed every 36 months after the original qualification approval date. The qualifying activity shall perform or witness the tests specified in Table III. Unless otherwise specified in Table III, sequential testing is required. The qualifying activity may shift the reporting date to accommodate testing schedules, but no later than 18 months from the retention of qualification due date.

4.3.3.1 No production certification. If no production of the qualified products has occurred for the entire reporting period, the supplier shall provide certification to the qualifying activity that no changes in the product materials, design, manufacturing processes,

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or site of production have occurred since qualification approval. Certification for more than one reporting period is not permitted. The supplier shall contact the qualifying activity to determine the conditions for certification. The supplier shall submit the certification on DD Form 1718 provided by the qualifying activity.

4.4 Conformance inspection.

4.4.1 Inspection of product for delivery. Inspection of product for delivery shall consist of conformance inspection tests specified in Table VIII.

4.4.1.1 Inspection lot. An inspection lot shall consist of assemblies (fastener and lacing strap) for a particular size produced under essentially the same manufacturing conditions and presented for inspection at one time.

4.4.1.2 Sampling plan. Inline statistical sampling shall be conducted during manufacturing at a rate of 1 per 1000 ±5 percent samples. For acceptance of the lot, there shall be zero occurrences of defects.

4.5 Examinations.

4.5.1 Visual and dimensional examination. Tie-down straps shall be examined to verify that the materials, design, construction, and physical dimensions are in accordance with this specification. Dimensions shall be measured with calibrated devices using standard laboratory methodology.

4.5.2 Examination of product (qualification inspection). Tie-down straps shall be carefully examined to determine compliance with the requirements of this specification with respect to workmanship, product identification, and dimensions as specified in 3.6, 3.7.9, 3.7.10 and 3.7.11.

4.5.3 Examination of product (conformance inspection). For conformance inspection, examination of product shall consist of the inspections specified in Table VIII.

4.6 Test methods.

4.6.1 Tensile strength (see 3.7.1). The test sample shall be securely attached and positioned on the tensile test fixture as shown on Figure 3. The fixture shall be positioned in a tensile testing machine and sufficient force applied to break the strap or, when applicable, to release the locking device or separate the strap from the locking device. The jaws of the tensile test machine shall move apart at a speed of one inch per minute

4.6.2 Thermal shock (see 3.7.2). The tie-down straps shall be mounted on a wire bundle and subjected to MIL-STD-202-107, Test Condition C. The wire bundle shall contain M22759/16-XX-X wire size 12 through 20 (e.g. M22759/16-12-9). The wire bundle shall be a minimum of 1.5 inches in diameter and permit the mounting of the tie-down straps once

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removed from the bundle to fit on the tensile strength fixture (see 4.6.4, Figure 3). The tie-down straps shall be spaced on the wire bundle so that no tie-down strap overlaps or touches another tie-down strap. The fastener and strap shall be inspected for cracks, distortions, breaks or releasing of the locking device, and then subjected to the tensile strength test after temperature exposure testing specified in 4.6.4.

a. Test Condition C – 5 cycles (Steps 1 through 4 are a complete cycle; see table 107-I, MIL-STD-202-107).

b. Using one thermal shock chamber or two temperature chambers, the chambers shall be cooled or pre-heated to the temperatures specified in steps 1 and 3 defined in 4.6.2.1 c and d.

c. Step 1 – Temperature: -65 +0, -5 °C; Time duration: 1 hour.

d. Step 3 – Temperature: +200 +5, -0 °C; Time duration: 1 hour.

e. Steps 2 and 4 – Temperature: Room ambient; Time duration: 2 minutes maximum (Steps 2 and 4 are the maximum times permitted to move the wire bundle to/from step 1 and 3 cold and hot temperatures; see table 107-I, MIL-STD-202-107).

4.6.3 Temperature life cycle (see 3.7.3). Thirty tie-down straps (10 single wrap, 10 double wraps and 10 triple wraps) shall be mounted on a wire bundle and subjected to temperature life cycle testing. The wire bundle shall contain M22759/10-XX-X wire size 12 through 20 (e.g. M22759/10-12-9). The wire bundle diameter shall be a minimum of 1.5 inches in diameter and permit the mounting of the tie-down straps once removed from the bundle to fit on the tensile strength fixture (see 4.6.3, Figure 3). The tie-down straps shall be spaced on the wire bundle so that no tie-down strap overlaps or touches another tie-down strap. The wire bundle with tie-down straps shall be placed in an oven for 1000 hours at 500 ±5 °F (260 ±3 °C). After completion, cool to room ambient temperature, while installed on the wire bundle, each fastener and strap shall be visually inspected for cracks, distortions, breaks or releasing of the locking device. The test wire bundle with the cable lacing fasteners shall then be subjected to the tensile strength test after temperature exposure test specified in 4.6.4.

4.6.4 Tensile strength after temperature exposure. The samples shall be prepared for the tensile strength test by removing the wire bundle from the test fixture then removing the center wire(s) from the bundle permitting the wire bundle to collapse. Slip the tie-down strap assembly off the collapsed bundle and remount to the test fixtures as shown on Figure 3. Perform the tensile strength test specified in 4.6.1.

4.6.5 Vibration test (see 3.7.4).

4.6.5.1 Test assembly. The tie-down strap configurations of single, double and triple wrap to be tested, shall be installed on the wire bundle diameter specified in Table IX with spacing and dimensions shown on Figure 4. For each tie-down strap configuration tested, a minimum of five wire bundles with 4 tie-down straps of each wrap configuration shall be installed. Wire bundle assemblies shall comprise a minimum of three M22759/16-XX-9 wires (12-24 AWG). The wire

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bundle may be taped on each end to prevent wire slippage during vibration. The wire bundle assemblies shall be installed on the vibration fixture using SAE AS21919WCJXX clamps or clamps capable of meeting vibration requirements of Table IX (see Table IX for clamp size and test wire bundle diameter).

4.6.5.2 Method of mounting. The assembly shall be installed flat on a vibration fixture and shaker table as shown on Figures 4, 5 and 6. Clamps shall be mounted flat on the vibration fixture with 10-32 UNF-3A bolts (160 ksi or higher strength) and torqued to 30 inch pounds ± 3 inch pounds. A torque stripe (indicator) shall be applied to the fasteners to ensure no rotation occurs. A control accelerometer shall be mounted on the vibration fixture. Response accelerometers shall be mounted on the vibration fixture in close proximity to both clamps as shown on Figure 5 and 6. Test assemblies may be mounted on a test fixture simultaneously provided that the capacity of the test equipment is not exceeded, and the test assemblies do not dynamically interact. If multiple test assemblies are tested simultaneously, a minimum of one test assembly size shall have the response accelerometer configuration shown on Figures 5 and 6.

4.6.5.3 Procedure.

a. Mount the test assemblies as specified in 4.6.5.1 and 4.6.5.2 on the fixture to the vibration machine.

b. Attach control accelerometers adjacent to the test assembly on the mounting fixture as shown on Figures 5 and 6.

c. Attach a triaxial accelerometer close to each clamp as shown on Figures 5 and 6.

4.6.5.4 Test method. Random vibration testing shall be conducted in accordance with Table IX in both the vertical and lateral axis as defined by Figures 4, 5, and 6. Test assemblies for each configuration of single, double or triple wrap, shall be tested once along a single axis. Wire bundles and mounting clamps may be used for more than one test of new test article components providing the bundles are in an acceptable condition to indicate abrasion from subsequent tests or to otherwise prevent any additional variation. Following each test, the test articles shall be inspected for cracks, distortions, breaks or releasing of the locking fastener.

4.6.5.5 Failure criteria (see 3.7.4). Tie-down straps shall not fail within the duration of the test. Visual damage to the strap is considered a failure. Visual damage includes cracks, breaks, or releasing of the lacing strap or locking fastener. Failure of the mounting clamp shall not be considered a device under test failure.

4.6.6 Fluid immersion (see 3.7.5). Individual components shall be installed on the tensile test fixture as specified on Figure 3. Two samples of each type and dash number shall be immersed for four hours in each of the fluids in Table V. Upon removal from the fluids, the components shall be placed on an absorbent material and shall remain for one hour in free air at room temperature. The components shall then pass the tensile strength test of 4.6.1 and product identification legibility test of 4.6.10.

4.6.7 Fuel immersion life (see 3.7.6). Each tie-down strap assembly shall be immersed in

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JP-8 (NATO F-34, see MIL-DTL-83133), JET A (NATO F-24, see ASTM D1655), or equivalent fuel for 1080 hours without cracking, distorting, breaking, separation of the fastener head from the lacing tape, or releasing of the locking device. Upon removal from the test fluid, the tie-down strap assembly shall be wiped dry and then remain for 1 hour in free air at room temperature. The tie-down strap assemblies shall then be subjected to the tensile strength test specified in 4.6.1.

4.6.8 Slip resistance test (see 3.7.7). Test shall be performed in a standard tensile testing machine with the specimen being pulled through a holding fixture that prevents the tie-down strap from passing through the fixture hole. Attach tie-down strap to test mandrel. The mandrel shall be gripped by the tensile machine jaws at one end and the other end of the mandrel shall be free to move through the holding fixture. Sufficient force shall be applied to cause the tie-down strap to slip on the mandrel. Test machine shall move at a rate of 1 in./min. Measure force required to make tie-down strap slip on the mandrel. Discontinue test when tie-down strap travels on the mandrel a distance of 0.25 inch. See figure 7 for test configuration.

4.6.8.1 Mandrel. The mandrel shall be constructed of polytetrafluoroethylene (PTFE) coated steel rod or tube. The tie-down strap shall be tested on mandrel sizes that are 1.0 inch outside diameter and 0.5 inch outside diameter.

4.6.9 Flammability test (see 3.7.8). A single wrapped tie-down strap shall be installed around the center of the mandrel. The mandrel assembly shall be clamped in the specimen holder of the test chamber at an angle of 60 degrees from horizontal angled away from the rear wall of the enclosure with the fastener head oriented towards the opening of the enclosure. The burner shall be adjusted to deliver an all-blue flame, 2 inches high, at a temperature of 1751 °F ±54 °F (955 °C ±30 °C), as measured with a thermocouple pyrometer. With the burner held at a 90 degree angle to the longitudinal axis of the mandrel and the long dimension of the flame spreader parallel to the longitudinal axis of the mandrel, the hot tip of the flame shall be applied to the tie-down strap so that the midpoint of the flame touches the fastener head on the assembled specimen. The period of flame application shall be 30 seconds. At the close of the application period, the flame shall be withdrawn and the duration of the after flame shall be recorded.

4.6.9.1 Apparatus. The flammability test chamber shall be approximately 1 foot square by 24 inches in height, and shall be open at top and front to provide required ventilation for combustion but prevent drafts. A mandrel of a 0.750-inch nominal diameter constructed of non-flammable metallic material shall be used to mount the test specimen. Means shall be provided in the chamber to hold the mandrel at an angle of 60 degrees from horizontal in a vertical plane parallel to and about 6 inches in front of the rear wall of the chamber. The test burner shall be an ASTM D5025 Bunsen type gas burner having a 0.250 inch inlet, a needle valve in the base for gas adjustment, a bore of 0.375 inch nominal, and a length of approximately 4 inches above the air inlets. The burner shall be fitted with a wing top flame spreader having a 0.0625-inch by 2-inch opening.

4.6.10 Product legibility (see 3.7.9). Product identification markings shall be visually examined for legibility. Ten X magnification assistance to the human eye is allowed.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The tie-down straps are intended for use in general wire and cable, including fiber optic and co-axial cable, bundle retention. The tie-down straps are designed to be used in high vibration, shock, highly corrosive, and high temperature environments unique to military applications. The tie-down strap may be used anywhere secondary support is required, but the user must verify the application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Tie-down strap part number (see 1.2).

c. Packaging requirements (see 5.1).

d. Minimum package identification information should include: Nomenclature, part identifying number, CAGE code, date manufactured, lot manufactured, the adjustable tie-down strap military specification number, and quantity (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No. 32554 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Commander, Naval Air Systems Command, 48298 Shaw Road, Building 1461, Patuxent River, MD 20670-1900 or QPL_Support@navy.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.3.1 Qualification inspection process. At the initiation of the qualification process, prior to

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testing, prospective suppliers must forward a written request for qualification to the qualifying activity (see 6.3). The qualifying activity will respond in writing with a formal "letter of authorization" providing detailed instructions for the submission of product samples. Testing cannot begin until the supplier has received this authorization letter. The qualification process is further described in the SD-6 "Provisions Governing Qualification." Copies of this document are available online at https://assist.dla.mil.

6.3.2 QPL supplemental information. The qualifying activity provides a summarized list of all qualified sources on a publicly accessible website http://www.navair.navy.mil/qpl/. The summary includes but is not limited to the supplier approved part number and related specification part number, a dedicated approval reference number, a supplier location where purchases may be requested and the manufacturing location of the component.

6.3.3 Qualification by similarity. The Qualifying Activity may qualify tie-down straps with the same lacing material and color (see Table I), but different lengths (see Table II) by similarity.

6.4 Installation of MIL-DTL-32554 tie-down straps. Use of the M32555/01-01 tool is recommended to ensure that the tie-down strap is correctly tensioned and not over tightened around the wire bundle. The fastener is designed to lock at a set minimum tension. Additional tensioning beyond the minimum does not improve the holding capability of the tie-down strap and may damage the wire bundle. The tie-down strap may be installed using multiple wraps of the strap around the wire bundle to achieve increased tensile strength as required (see Table IV). Additional wraps may provide additional support for larger bundles and reduce compression of coaxial and fiber optic cables by increasing the surface area of the strap around the cable/wire harness. The process of applying traditional lacing tape (e.g. A-A-52084-C-2) results in two wraps of the lacing tape around the cable/wire bundle to form a clove hitch knot to complete the installation.

6.5 Wire bundles containing coaxial and fiber optic cables. Installation of adjustable plastic straps (see SAE AS23190) or lacing tape A-A-52080 through A-A-52084 (see SAE AS50881) onto wire bundles containing fiber optic cables, Low Voltage Differential Signaling (LVDS) data cables, and/or coaxial cables may induce signal degradation. See Cognizant Engineering Authority for guidance. These cables may be moved internal to the wire bundle so direct pressure from the tie-down strap is reduced.

6.6 Aircraft fuel tank installation. The tie-down strap has been tested for aviation fuel compatibility and may be used for wire bundle secondary support or spot tie of wire/wire harnesses in fuel tanks or fuel vapor environments.

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes."

6.8 Subject term (key word) listing.

Clamping cables	Fastening device
Fastening aircraft cables	Wire harness

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TABLE I. Tie-down strap material.

Lacing Tape Material Designator	Lacing Tape Material	Lacing Tape color 1/
-1	A-A-52084-C-2	-XXX

1/ See color code in A-A-52084, Table 2.

TABLE II. Tie-down strap dimensions.

Length Designator	Strap length L (in.) +0.35, -0.00	Fastener Length A (in.) Max	Fastener Width B (in.) Max	Fastener Height C (in.) Max	Weight (oz.) Max
06	6.00	0.300	0.250	0.200	0.011
10	10.00	0.300	0.250	0.200	0.013
18	18.00	0.300	0.250	0.200	0.020
24	24.00	0.300	0.250	0.200	0.024
44	44.00	0.300	0.250	0.200	0.040

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TABLE III. Qualification inspections.

Examination or Test	Requirement paragraph	Test paragraph (in sequence)	Number of samples
Visual examination 1/	3.6, 3.7.9, 3.7.10, 3.7.11	4.5.1, 4.5.2	10
Tensile strength 1/	3.7.1	4.6.1	10
Thermal shock	3.7.2	4.6.2	10
Tensile strength after temperature exposure	See Table IV	4.6.4	
Temperature life cycle	3.7.3	4.6.3	10
Tensile strength after temperature exposure	See Table IV	4.6.4	
Vibration 1/, 2/	3.7.4	4.6.5	
Fluid immersion	3.7.5	4.6.6	10
Fuel immersion	3.7.6	4.6.7	10
Slip resistance 1/	3.7.7	4.6.8	10
Flammability	3.7.8	4.6.9	10
Product legibility 1/	3.7.9	4.6.10	10

1/ Retention of qualification inspections to be conducted by the qualifying activity.
2/ Retention of qualification inspection test repeated every other retention cycle.

TABLE IV. Minimum tensile strength. 1/

No. of wraps around mandrel	Tensile strength (lbf)
1	55
2	110
3	165

1/ Tensile strength limits applicable for pre and post temperature shock and temperature life cycle testing.

TABLE V. Fluids.

Test fluid	Test temperature (°F)	Immersion time (hours)
MIL-PRF-5606, MIL-PRF-83282, or MIL-PRF-87257 Hydraulic Fluid	118-122	4
MIL-PRF-7808, Lubricating Oil		
TT-I-735, Isopropyl Alcohol	68-77	

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TABLE VI. Minimum tensile strength after fluid and fuel immersion.

No. of Wraps Around Mandrel	Hydraulic Fluid (lbf)	Lubricating Oil (lbf)	Isopropyl Alcohol (lbf)	Fuel (lbf) 1/
1	55	55	55	-
2	110	110	110	-
3	165	165	165	135

1/ Tensile strength after fuel immersion life test. Only 3 wraps are recommended for fuel tanks for safety and strength.

TABLE VII. Minimum slip resistance.

No. of Wraps Around Mandrel	1.000 inch mandrel (lbf)	0.500 inch Mandrel (lbf)
1	1.30	2.23
2	2.55	3.18
3	2.24	3.43

TABLE VIII. Conformance inspection.

Examination and Tests	Requirement Paragraph	Test Paragraph	Sampling plan
Visual examination	3.6, 3.7.9, 3.7.10, 3.7.11	4.5.1	4.4.1.2
Tensile Strength 1/	3.7.1	4.6.1	
Product Legibility	3.7.9	4.6.10	

1/ Testing of 6-inch length tie down strap not required.

TABLE IX. Random vibration profiles by bundle diameter.

Break points for random vibration profile						
Clamp Size	Bundle Diameter (inches)	Overall Acceleration [g_{RMS}]	20 Hz Breakpoint [g^2/Hz]	500 Hz Breakpoint [g^2/Hz]	800 Hz Breakpoint [g^2/Hz]	Duration [minutes]
Random vibration vertical						
-16	1	18	0.604	0.604	0.0074	17
-24	1.5	8	0.119	0.119	0.0014	45
-32	2	7	0.091	0.091	0.0011	24
Random vibration lateral						
-16	1	18	0.604	0.604	0.0074	28
-24	1.5	12	0.266	0.266	0.0030	20
-32	2	8	0.119	0.119	0.0014	28

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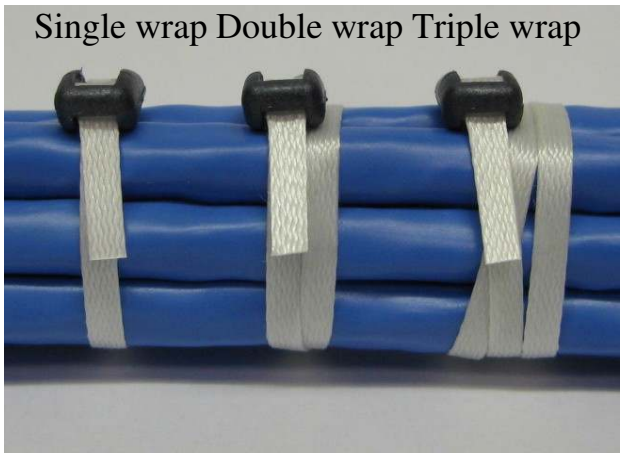
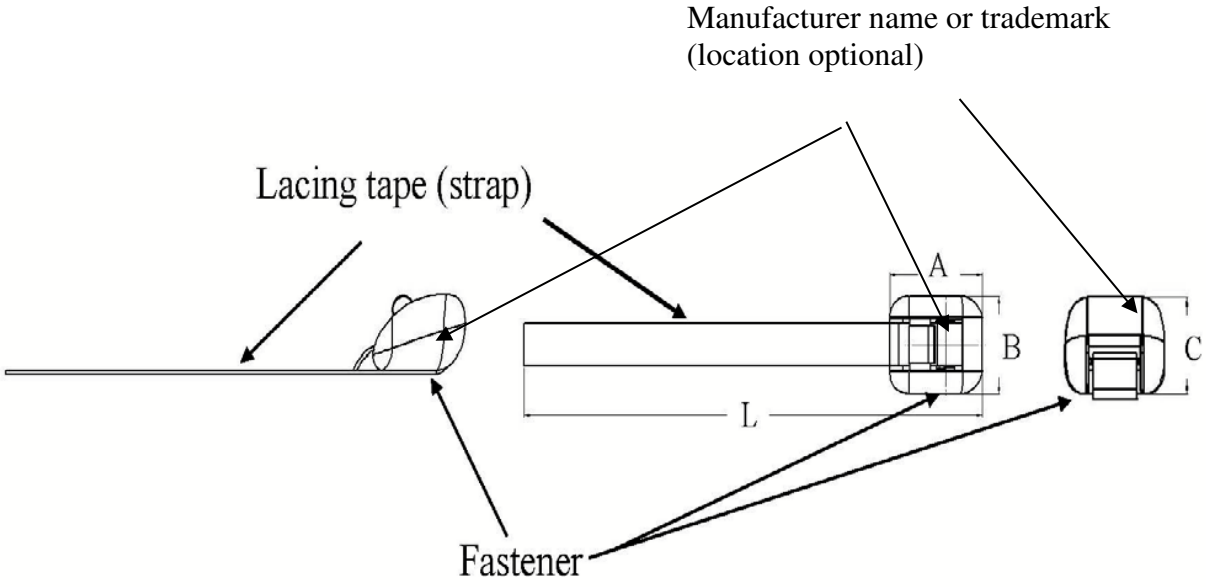


FIGURE 1. Installed tie-down strap.



NOTE: See Table II for tie-down strap dimensions

FIGURE 2. Tie-down strap assembly.

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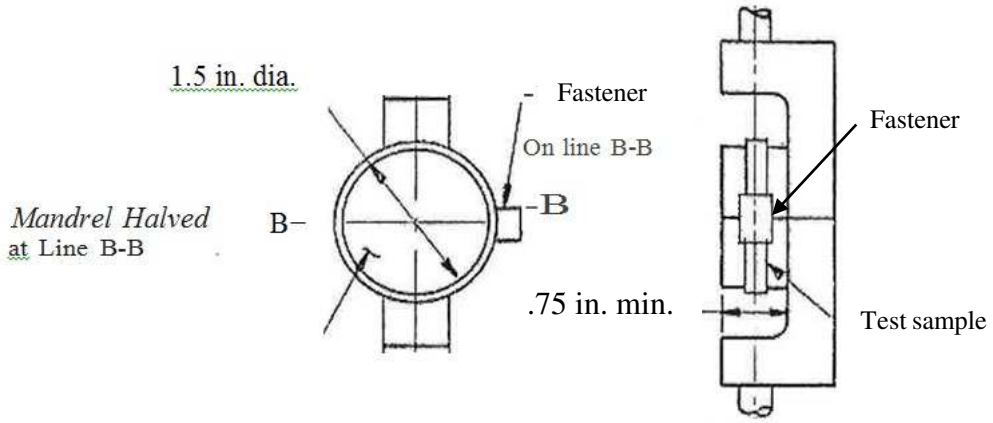
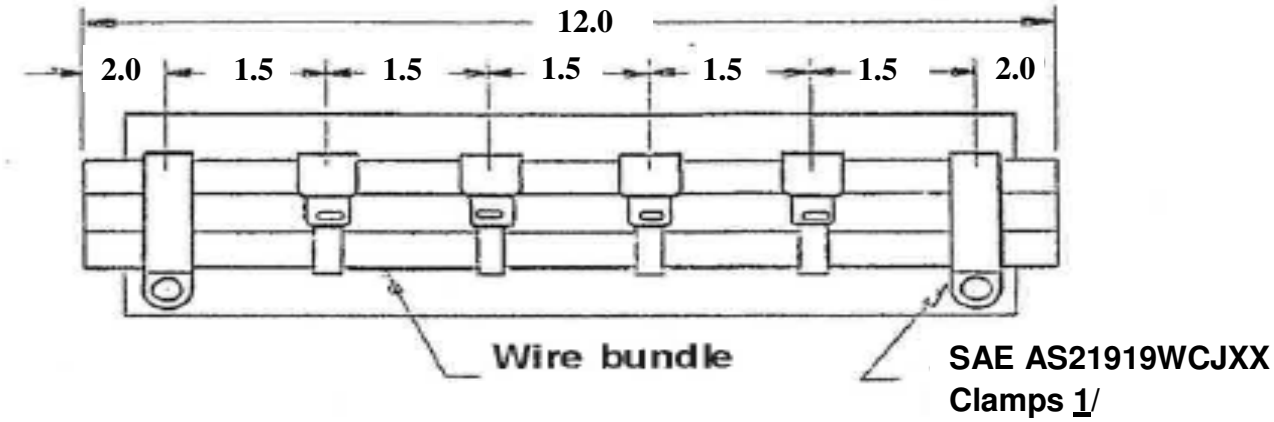


FIGURE 3. Tensile test fixture for strap.



1/ Two SAE AS21919 clamps or clamps capable of meeting vibration requirements (see 4.6.5.1) are required, XX size to fit the bundle diameter (see Table IX for clamp size and test wire bundle diameter).

FIGURE 4. Vibration tie-down strap spacing and test fixture.

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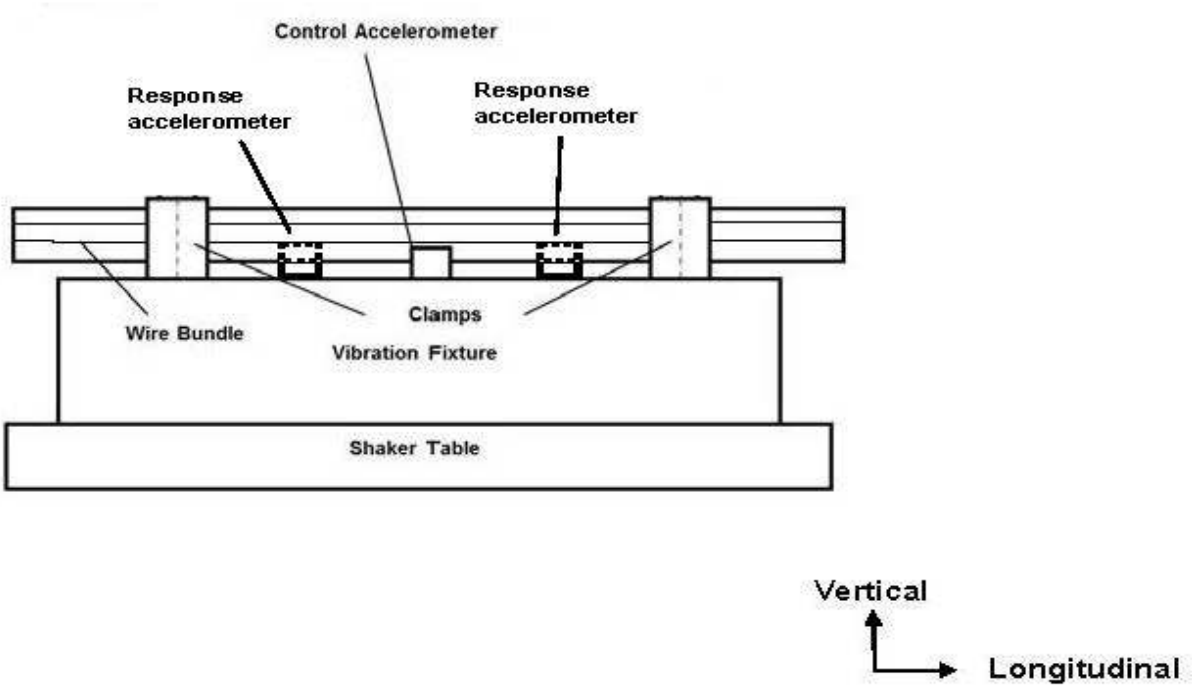


FIGURE 5. Vibration test configuration (side view).

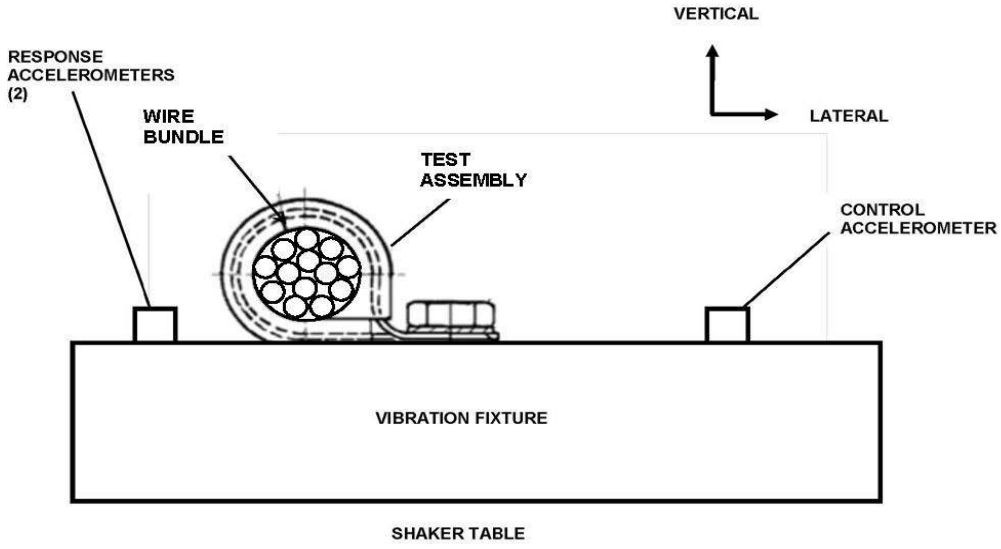


FIGURE 6. Vibration test configuration (end view).

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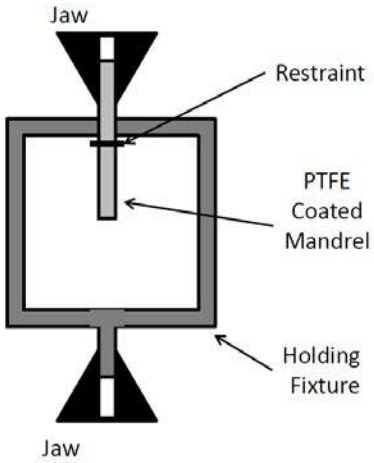


FIGURE 7. Slip resistance test configuration.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Preparing activity:
Navy-AS

Project 5975-2020-002

Review activities:
Army - AV, MI
Navy - EC, OS, SH
Air Force - 11, 20
DLA - GS

Industry association:
SAE AE-8C2

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